

Multiple Choice (20 marks)

1. A person has a clean yard at the beginning of a season. As the seasons go on and the trees in the yard grow,
 - (a) more trees are indoors
 - (b) the yard needs to be raked more
 - (c) some trees are inside
 - (d) fewer leaves are in the yard
2. Which of the following does authorization aim to accomplish?
 - (a) Restrict what operations/data the user can access
 - (b) Determine if the user is an attacker
 - (c) Flag the user if he/she misbehaves
 - (d) Determine who the user is
3. Which of the following is true of calcium metabolism?
 - (a) Calcitonin causes a rise in plasma calcium
 - (b) Primary hyperparathyroidism is usually asymptomatic
 - (c) Vitamin D is secreted by the parathyroid glands
 - (d) Oliguria is a symptom of hypercalcaemia
4. A single-electron atom has the electron in the $l = 2$ state. The number of allowed values of the quantum number m_l is
 - (a) 5
 - (b) 4
 - (c) 3
 - (d) 2
5. Jackrabbits live in very hot environments, but are able to stay cool because their long ears help
 - (a) releasing warmth
 - (b) sing songs
 - (c) make eggs
 - (d) change tires
6. Which of the following is a typical example of a unicellular organism?

- (a) earthworm
 - (b) bacteria
 - (c) fungi
 - (d) green algae
7. A ferromagnetic metal, such as in paper clips, can stick to other paper clips without hooking together
- (a) when outside
 - (b) when iron touches
 - (c) when floating
 - (d) when burned
8. A landslide is likely to occur on a slope as
- (a) there is less vegetation
 - (b) downward movement is facilitated
 - (c) there is more debris
 - (d) it means more shifting occurs there.
9. Which is the best buy?
- (a) 3 bottles of juice for \$6.72
 - (b) 11 bottles of juice for \$24.86
 - (c) 9 bottles of juice for \$20.25
 - (d) 16 bottles of juice for \$35.68
10. Which planet has the longest planetary year?
- (a) Earth
 - (b) Venus
 - (c) Jupiter
 - (d) Neptune

True / False (20 marks)

Answer True or False and justify briefly.

1. True or False: The correct answer to 'How many chromosomes are in a human sperm cell?' is '23'.

2. True or False: The correct answer to 'Roger poured water over a pile of sand. Some of the sand washed away. This process is similar to which of the following?' is 'The eruption of a volcano'.
3. True or False: The correct answer to 'Hawkmoths are insects that are similar in appearance and behavior to hummingbirds. Which of the following is LEAST valid?' is 'These organisms have analogous structures.'.
4. True or False: The correct answer to 'Who famously explicated the science of dropping something?' is 'Rutherford'.
5. True or False: The correct answer to 'If an object undergoes chemical change then that object will have new chemical properties, such as' is 'a soda with lime in it'.
6. True or False: The correct answer to 'A cell that is low in water will most likely immediately lose the ability to' is 'remain rigid.'.
7. True or False: The correct answer to 'What primarily drives the hydrologic cycle to produce clouds?' is 'energy from a gun'.
8. True or False: The correct answer to 'Andy lives in the southern hemisphere. What season does he most likely experience in August?' is 'winter'.
9. True or False: The correct answer to 'When trees are harvested, the trees' is 'are smaller and less'.
10. True or False: The correct answer to 'The first commercial radio station was located in what city?' is 'Austin'.

End of Paper